IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Methodologies:**
  - Collected data via SpaceX API and Wikipedia scraping;
  - Wrangled for cleanliness and ML compatibility;
  - Conducted EDA with visualizations and SQL queries;
  - Created interactive Folium maps and Plotly Dash dashboards;
  - Built and tuned classification models (Logistic Regression, SVM, Decision Tree, KNN).
- **Key Results:**
  - Launch success rates improved from 0% in 2013 to 100% in 2020;
  - Highest success in 2000-4000 kg payload range;
  - SSO orbits at 100% success;
  - KSC LC-39A leads with 76.9% rate;
  - Models achieved 83% test accuracy on test data, with Decision Tree and SVM excelling in cross-validation.
- **Implications:** Falcon 9 reusability reduces costs to $62M per launch vs. competitors' $165M. SpaceY can leverage insights for low-risk bids, focusing on optimal payloads, orbits, and coastal sites.

Introduction

- **Background:** SpaceX's Falcon 9 achieves cost savings through reusable first stages, with successful landings enabling multiple flights per booster.

- **Problem:** Predict first-stage landing outcomes based on variables like payload mass, orbit type, and launch site to inform bidding strategies for competitor SpaceY.

- **Data:** 90 launches (2010-2021) from SpaceX REST API (launch details) and Wikipedia scraping (booster versions and outcomes); key features: payload mass (mean 3,066 kg), 11 orbit types, 4 U.S. coastal sites.

Section 1

Methodology

Methodology

Executive Summary

- **Data Collection:**
  - API requests for structured launch data;
  - Web scraping for outcomes;
  - Merged datasets on flight number.
- **Data Wrangling:**
  - Handled missing values (e.g., payload imputed with mean 3,066 kg);
  - Binary encoded success (1/0);
  - One-hot encoded categoricals like sites and orbits.
- **EDA:**
  - Visualized trends with Seaborn/Matplotlib;
  - Aggregated insights via SQL on SQLite.
- **Interactive Analytics:**
  - Folium for geospatial mapping of sites and proximities;
  - Plotly Dash for dynamic success/payload explorations.
- **Predictive Modeling:**
  - 80/20 train/test split;
  - Hyperparameter tuning with GridSearchCV;
  - Evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.

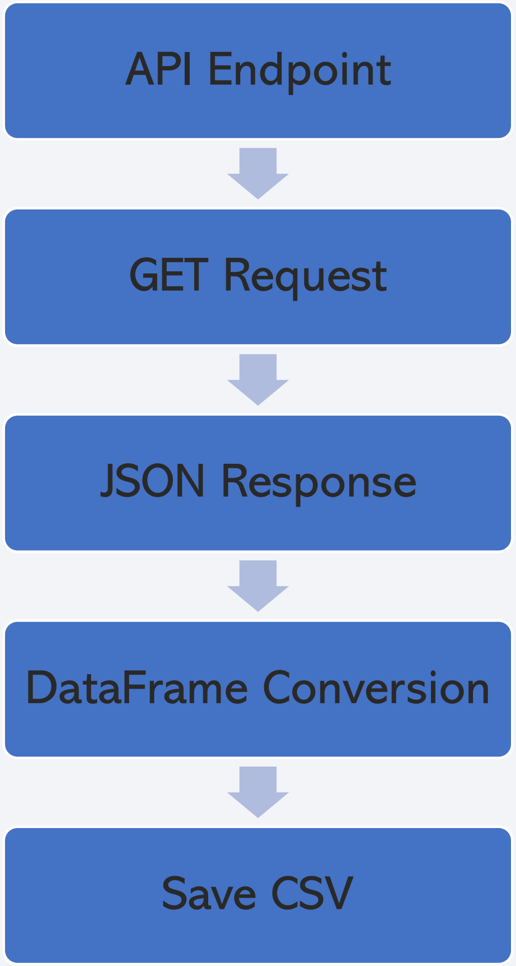
Data Collection

- **Process:**
  - Used requests library for API JSON fetches;
  - BeautifulSoup for Wikipedia table parsing.
  - Flowchart:

    Start → API Get Launches → Parse to DF → Scrape Wiki → Merge on Flight Number → End.
- **Key:**
  - RESTful integration;
  - HTML extraction;
  - Data alignment for 90 complete records.

Data Collection – SpaceX API

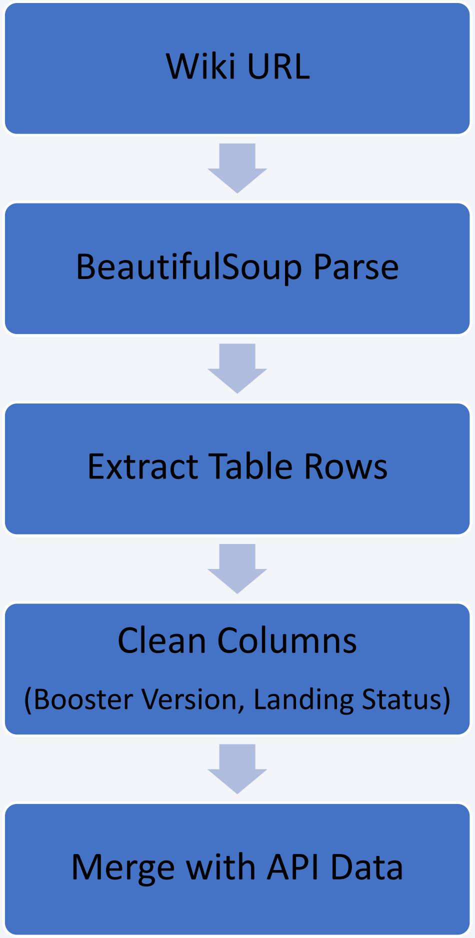
- **Description:**
  - Queried v4/launches for flight_number, launch_site, payload_mass_kg, orbit, core landing outcomes;
  - Processed 90 entries.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/1.1_SpaceX_Data_Collection_API.ipynb



Data Collection - Scraping

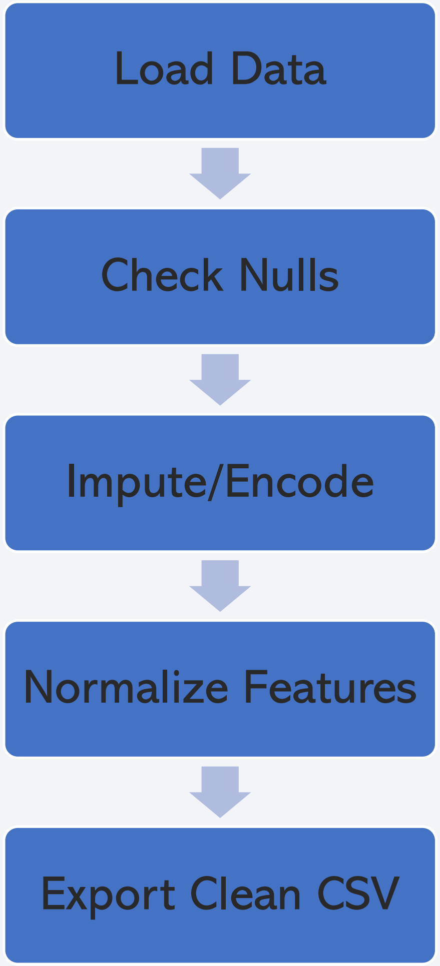
- **Description:**
  - Targeted Falcon 9 history table;
  - Converted outcomes to binary success/failure.

- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/1.2_Web_Scraping.ipynb



Data Wrangling

- **Process:**
  - Dropped irrelevant columns;
  - Imputed payload NaNs with mean (3066 kg);
  - One-hot encoded launch sites (4 unique) and orbits (11 unique);
  - Binary outcomes.
- **Key Phrases:**
  - Feature engineering for ML;
  - Added class imbalances.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/1.3_SpaceX_Data_Wrangling.ipynb



EDA with Data Visualization

- **Charts Plotted:**
  - Scatter (Flight Number vs. Launch Site) for site evolution;
  - Scatter (Payload vs. Launch Site) for capacity trends;
  - Bar (Success Rate vs. Orbit) showing SSO at 100%;
  - Scatter (Flight Number vs. Orbit);
  - Scatter (Payload vs. Orbit);
  - Line (Yearly Success Rate) from 0% in 2013 to 100% in 2020.
- **Rationale:**
  - Revealed correlations like higher success with lower payloads;
  - Informed feature selection.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/2.2_EDA_Data_Visualization.ipynb

EDA with SQL

- **Queries Performed:**
  - Unique launch sites;
  - Sites beginning with 'CCA' sites (5 records);
  - Total NASA payload (45,596 kg);
  - Avg payload for F9 v1.1 (2,928 kg);
  - First ground landing date (2015-12-22);
  - Boosters with drone success and 4000-6000 kg payload (F9 FT B1022, etc.);
  - Success/failure counts (100/1);
  - Max payload boosters (F9 B5 B1048.4, etc.);
  - 2015 failures (F9 v1.1 B1012, etc.);
  - Ranked outcomes 2010-2017 (No attempt 10, etc.).

- **GitHub URL:** https://github.com/faansing/IBM-Data-Science/blob/main/2.1_EDA_SQL.ipynb

Build an Interactive Map with Folium

- **Map Objects:**
  - Site markers color-coded by success;
  - Circles for proximity radii;
  - Lines to coastlines/railways/highways/cities with calculated distances (e.g., 0.86 km to coast).
- **Rationale:**
  - Coastal site enable over-water paths;
  - proximities aid logistics/safety.

- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/3.1_Launch_Site_Location.ipynb

Build a Dashboard with Plotly Dash

- **Plots:**
  - Pie chart for site success rates;
  - Scatter for payload vs. outcome with range slider and site dropdown;
  - Booster version color-coding.
- **Rationale:**
  - Enabled exploration of patterns like 76.9% success at KSC LC-39A and optimal 2000-4000 kg range.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/3.2_SpaceX_Dash_App.py

Predictive Analysis (Classification)

- **Process:**
  - Feature scaling;
  - Model training and tuning via GridSearchCV;
  - All models at 83% test accuracy;
  - Decision Tree/SVM best with balanced precision/recall.
- **Flowchart:**
  - Preprocess Data → Split Train/Test → Train Models → Tune Hyperparams → Evaluate Metrics → Select Best.
- **GitHub URL:**
  - https://github.com/faansing/IBM-Data-Science/blob/main/4.1_SpaceX_Machine_Learning_Prediction.ipynb

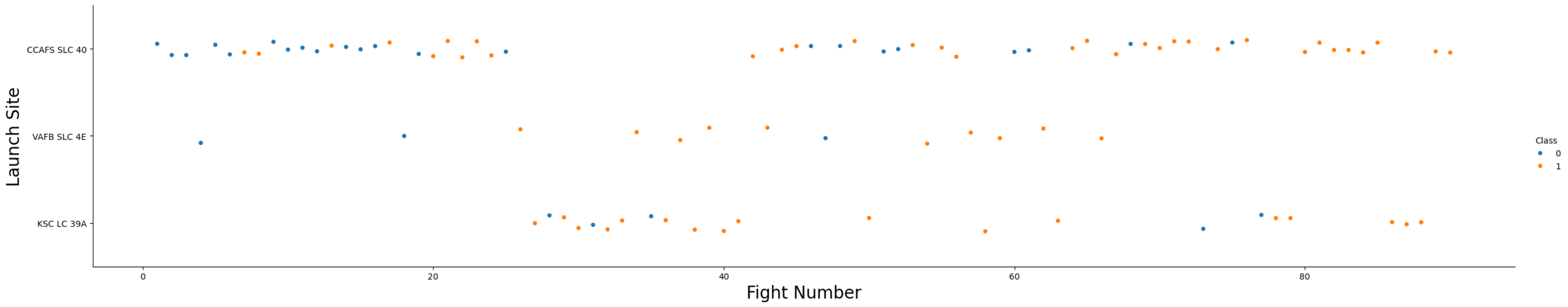
Results

- **EDA Visuals:**
  - Success rates rose over time;
  - Lower payloads (< 4000 kg) and orbits like SSO correlate with higher success.
- **Interactive Demos:**
  - Maps show coastal advantages;
  - Dashboards reveal booster FT excels in 2000-4000 kg.
- **Predictive Results:**
  - 83% accuracy across models;
  - Key features: orbit (0.35 importance), payload (0.25).

Section 2

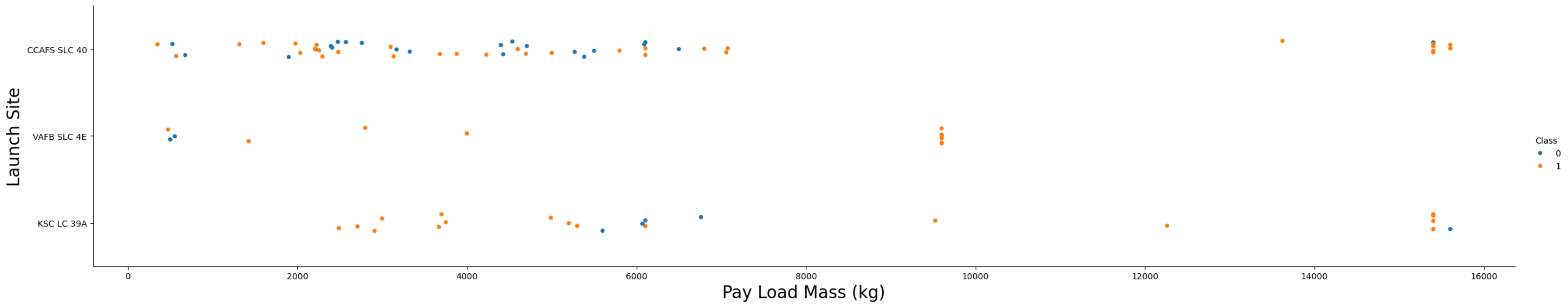
Insights drawn from EDA

Flight Number vs. Launch Site



- Explanation:
  - Early flights at CCAFS had varied outcomes;
  - KSC LC-39A shows consistent successes after flight 30, indicating operational maturity.

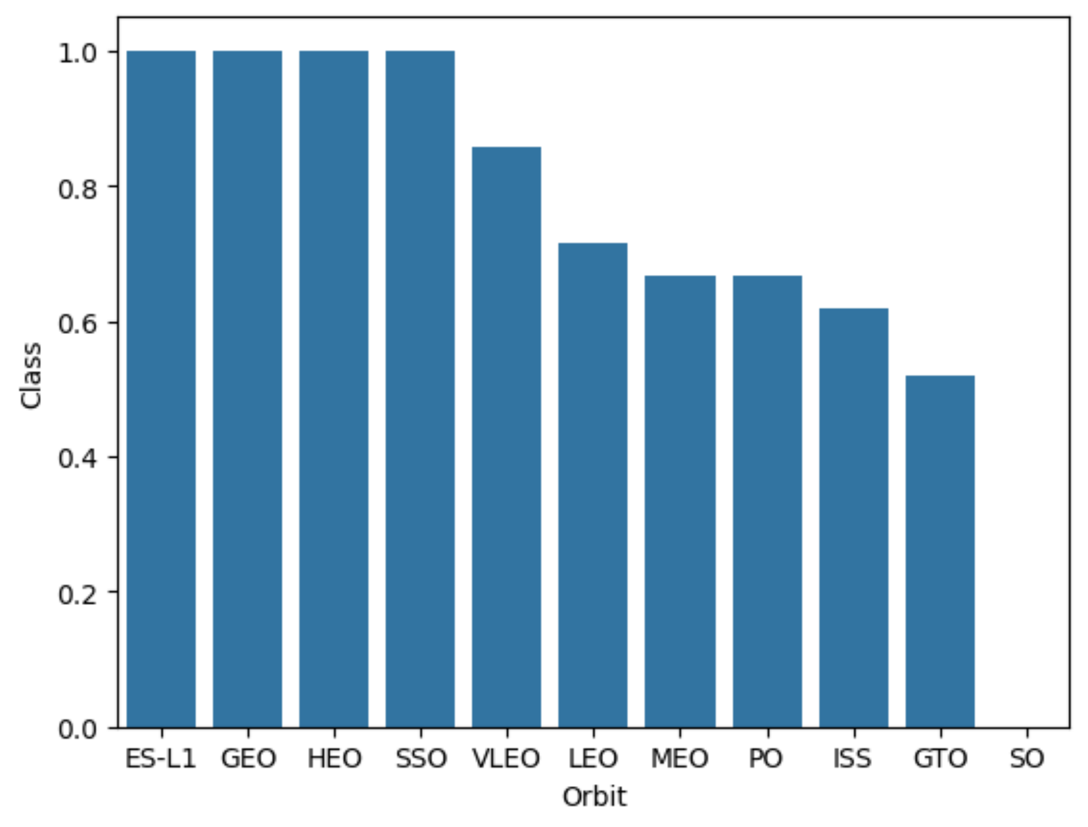
Payload vs. Launch Site



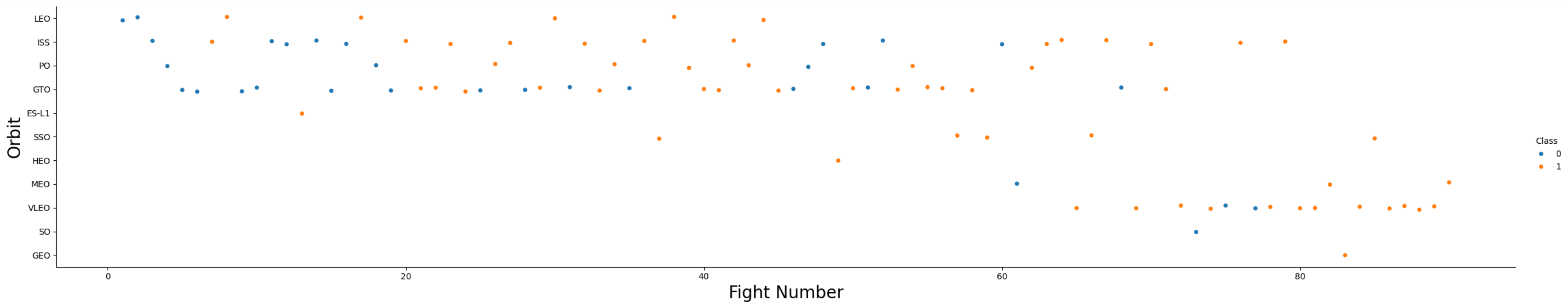
- Explanations:
  - VAFB SLC-4E handles lighter payloads (<5000 kg) with high success;
  - Heavier loads at CCAFS correlate with failures due to demanding orbits.

Success Rate vs. Orbit Type

- Explanation:
  - ES-L1, GEO, HEO and SSO at 100%;
  - GTO at ~50% due to higher energy requirements – critical for feature prioritization.

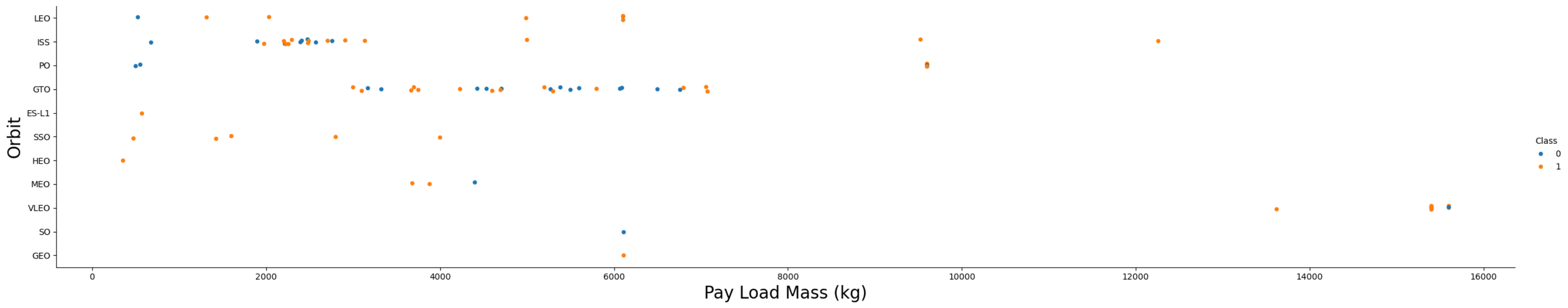


Flight Number vs. Orbit Type



- Explanation:
  - Shift to VLEO in later flights with improved success rates, reflecting technological advancements.

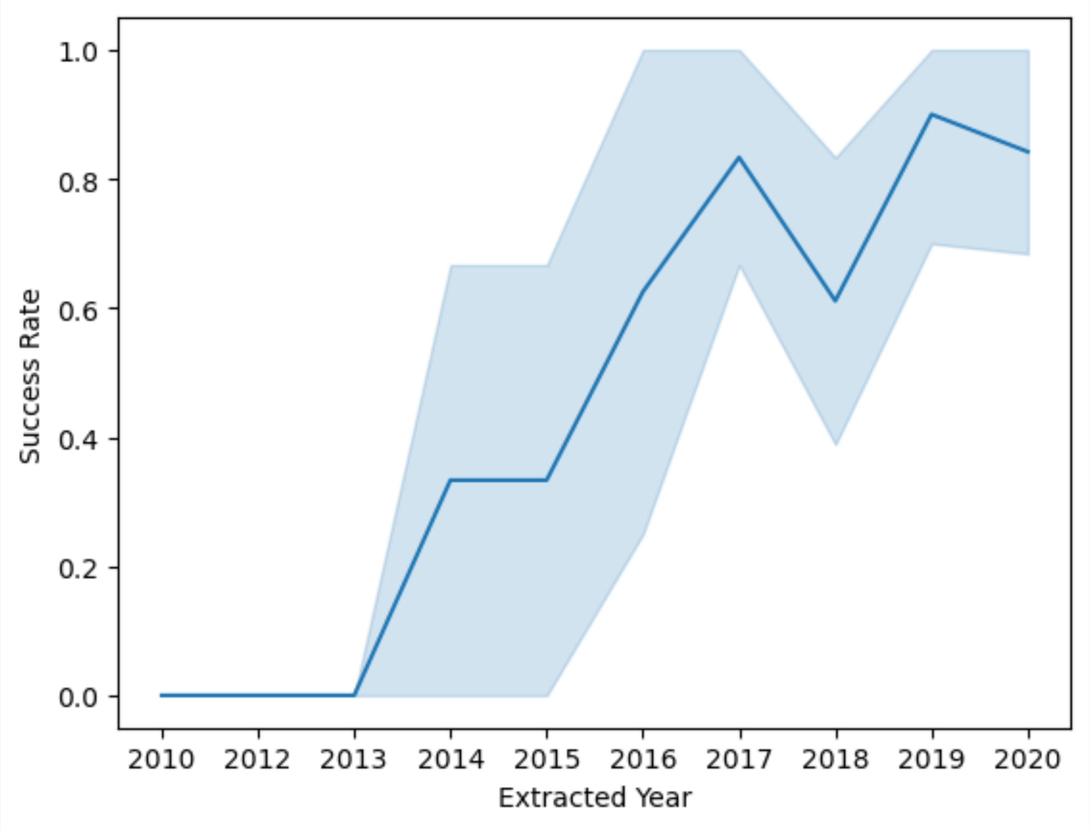
Payload vs. Orbit Type



- Explanation:
  - Optimal success in 2000-4000 kg range;
  - Heavier payloads (>6000 kg) in GTO show more failures.

Launch Success Yearly Trend

- Explanation:
  - Steady rise from 0% in 2013 to 100% in 2020, driven by landing technology refinements.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

In [11]:
```
## Downgrad `prettytable` to version 3.5.0 to ensure compatibility with ipython-sql-0.5.0

!pip install prettytable==3.5.0
```

```
Requirement already satisfied: prettytable==3.5.0 in /opt/conda/lib/python3.12/site-packages (3.5.0)
Requirement already satisfied: wcwidth in /opt/conda/lib/python3.12/site-packages (from prettytable==3.5.0) (0.2.1
3)
```

In [12]:
```
%sql SELECT DISTINCT launch_site FROM SPACEXTBL
```

```
 * sqlite:///my_data1.db
Done.
```

Out[12]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Explanation:
    - Four unique coastal U.S. sites optimized for equatorial and polar orbits.

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

In [13]:
```
%sql SELECT * FROM SPACEXTBL WHERE launch_site LIKE 'CCA%' LIMIT 5
```

```
 * sqlite:///my_data1.db
Done.
```

Out[13]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (p
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (p
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	N
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	N
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	N

- Explanation:
  - Filters early Cape Canaveral launches for site-specific trends.

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

In [14]:
```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTBL WHERE customer LIKE 'NASA (CRS)'
```

```
 * sqlite:///my_data1.db
Done.
```

Out[14]:

Total_Payload_Mass
45596

- Explanation:
    - Quantifies NASA's reliance on SpaceX for cargo missions.

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

In [15]:
```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1'

# 'F9 v1.1  B1003', 'F9 v1.1', 'F9 v1.1 B1011', 'F9 v1.1 B1010', 'F9 v1.1 B1012', 'F9 v1.1 B1013', 'F9 v1.1 B1014
```

```
 * sqlite:///my_data1.db
Done.
```

Out[15]:

Avg_Payload_Mass
2928.4

- Explanation:
    - Benchmark for early booster capabilities.

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

In [16]:
```
%sql SELECT MIN(Date) AS Date FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)'
```

```
 * sqlite:///my_data1.db
Done.
```

Out[16]:

Date
2015-12-22

- Explanation:
  - Key milestone in reusability history.

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

In [17]:
```
%sql SELECT Booster_Version FROM SPACEXTBL WHERE (Landing_Outcome = 'Success (drone ship)') AND (PAYLOAD_MASS__KG
```

```
 * sqlite:///my_data1.db
Done.
```

Out[17]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Explanation:
  - Identifies mid-range payload successes on ocean platforms.

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

In [18]:
```
%sql SELECT Mission_Outcome, COUNT(*) AS Count FROM SPACEXTBL GROUP BY Mission_Outcome ORDER BY Mission_Outcome


# %sql SELECT COUNT(Mission_Outcome) AS Sueecss_Outcome FROM SPACEXTBL WHERE Mission_Outcome LIKE 'Success%'
# %sql SELECT COUNT(Mission_Outcome) AS Failure_Outcome FROM SPACEXTBL WHERE Mission_Outcome LIKE 'Failure%'
```

```
 * sqlite:///my_data1.db
Done.
```

Out[18]:

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- Explanation:
  - Demonstrates overall mission reliability.

Boosters Carried Maximum Payload

Task 8

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

In [19]:
```
%sql SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ IN (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXT
```

```
 * sqlite:///my_data1.db
Done.
```

Out[19]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

- Explanation:
  - Highlights top heavy-lift boosters.

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

In [20]:
```
%sql SELECT SUBSTR(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE SUBST
```

```
 * sqlite:///my_data1.db
Done.
```

Out[20]:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Explanation:
  - Early drone ship landing challenges in 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

In [21]:

```
%sql SELECT Landing_Outcome, COUNT(Landing_Outcome) AS Count FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '
# %sql SELECT LANDING__OUTCOME FROM SPACEXTBL WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' ORDER BY DATE DESC
```

```
 * sqlite:///my_data1.db
Done.
```

Out[21]:

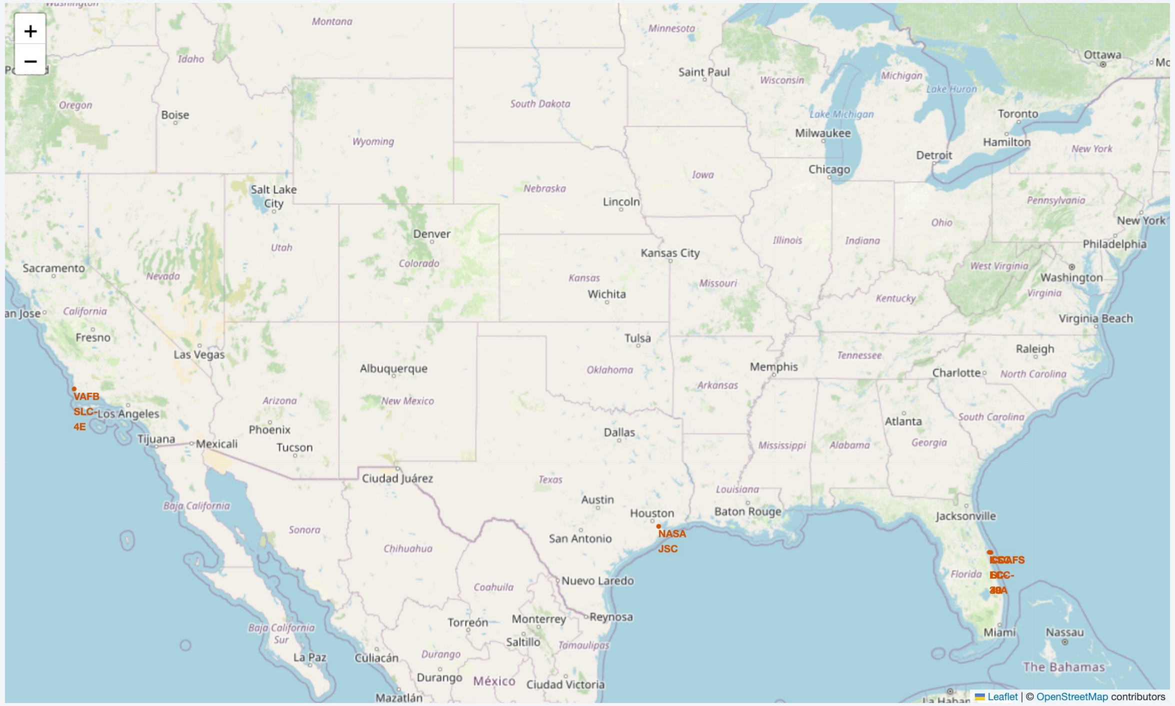
Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Explanation:
  - Shows progression from no attempts to successful landings.

Section 3

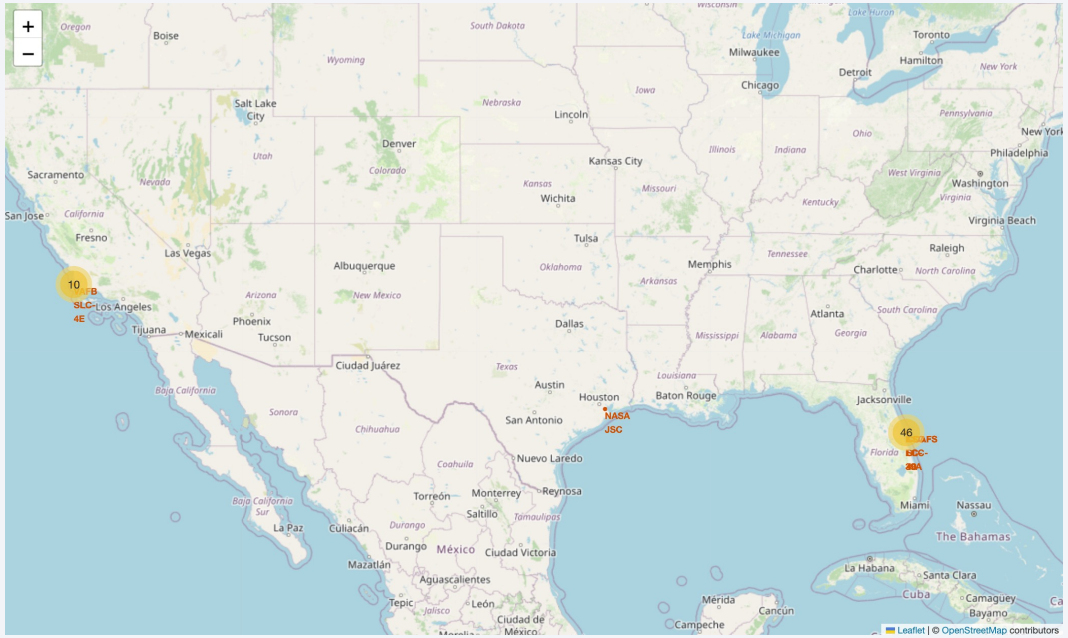
Launch Sites Proximities Analysis

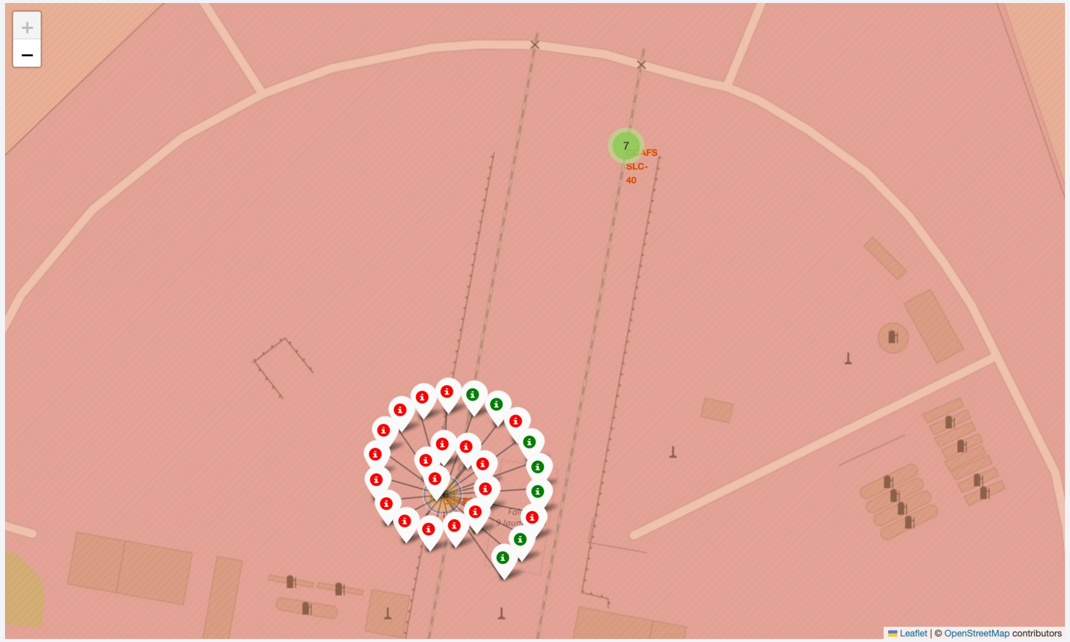
Folium Map: Global Launch Sites



- Explanation:
  - Sites clustered on U.S. coasts for over-water trajectories;
  - Eastern sites favor equatorial orbits.

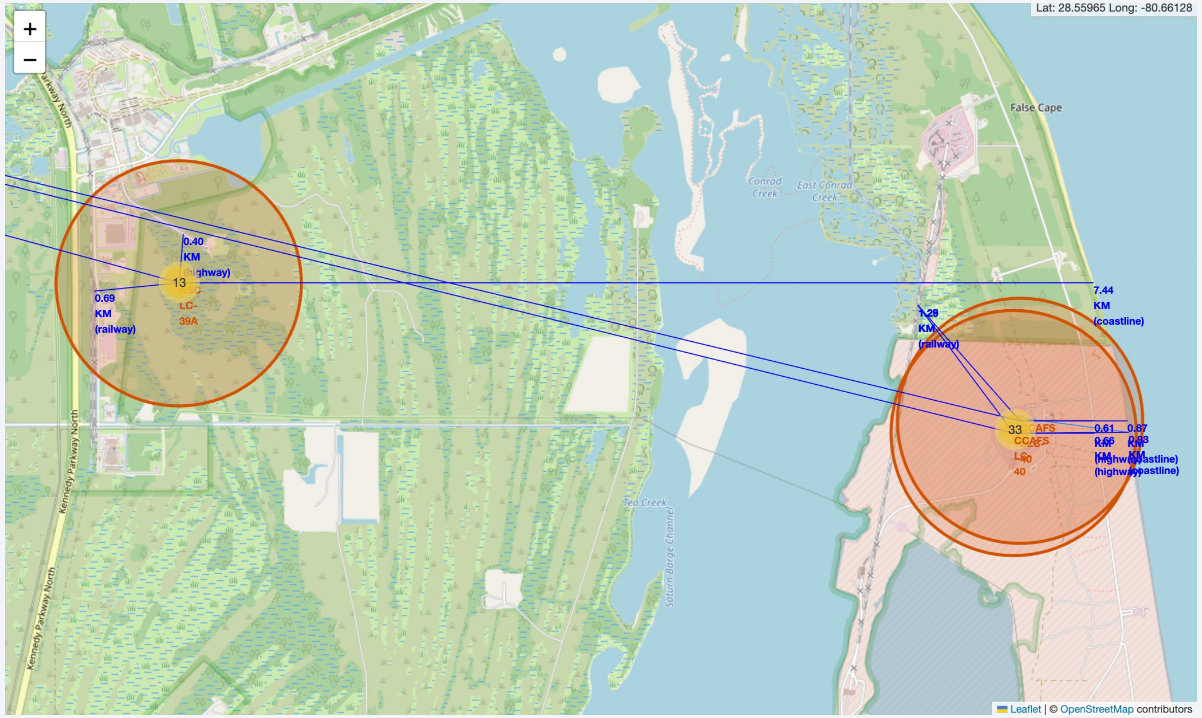
Folium Map: Launch Outcomes





- Explanation:
  - Green clusters near coasts indicate high success;
  - Red markers often offshore for drone ships.

Folium Map: Site Proximities

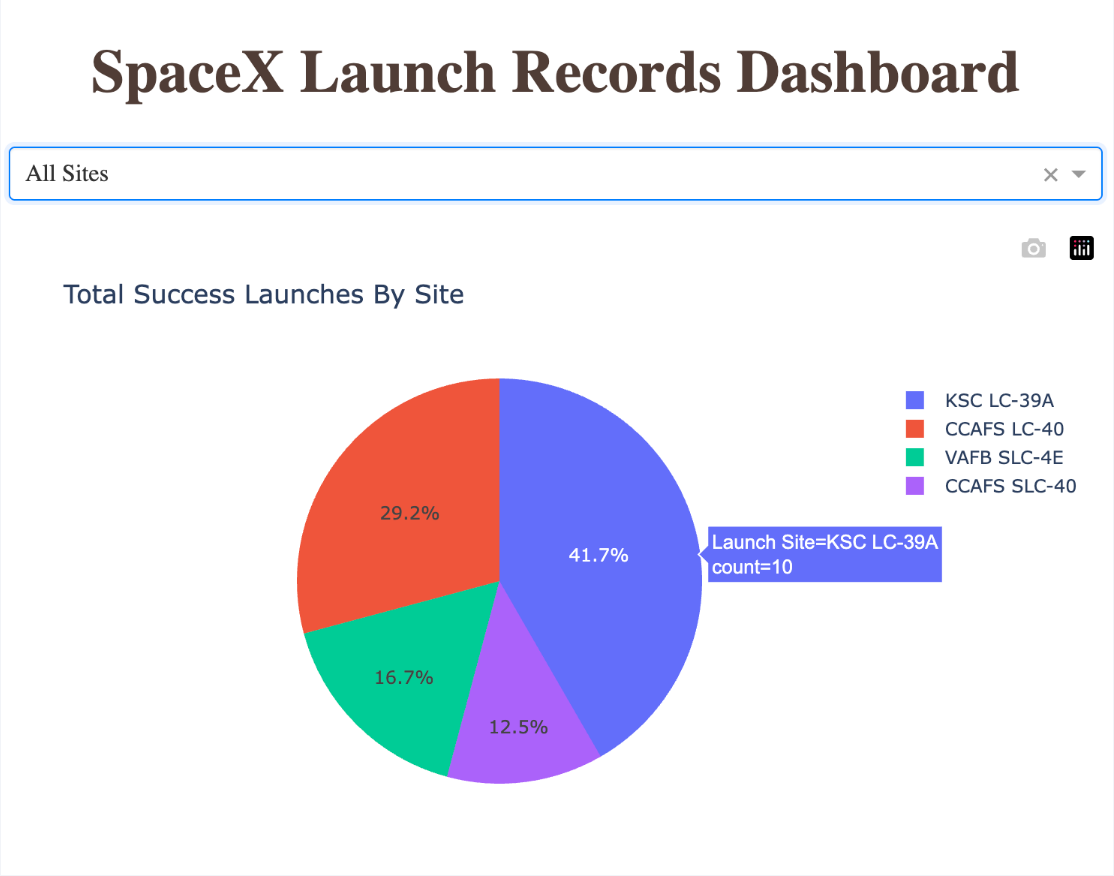


- Explanation:
  - Proximity to infrastructure supports logistics;
  - Distance from cities ensures safety.

Section 4

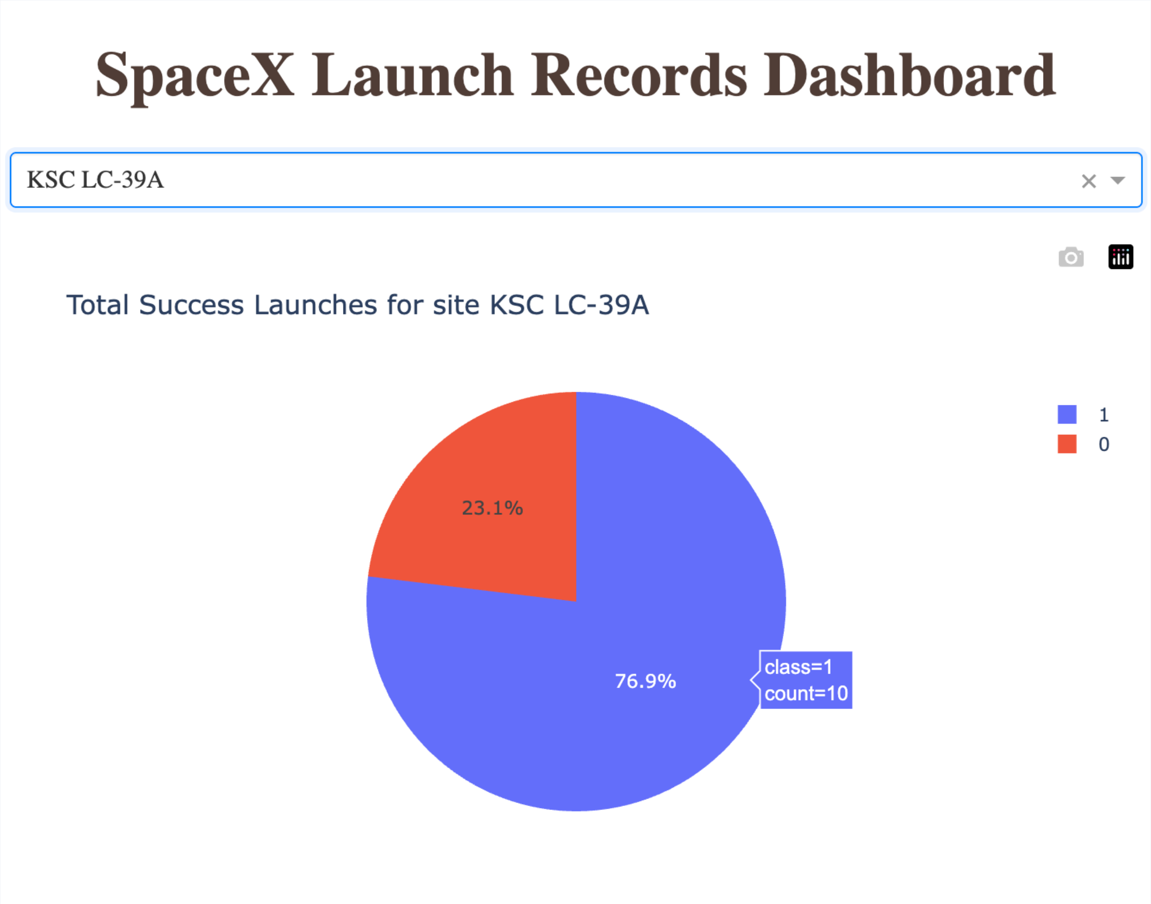
Build a Dashboard with Plotly Dash

Dashboard: Launch Success by Site



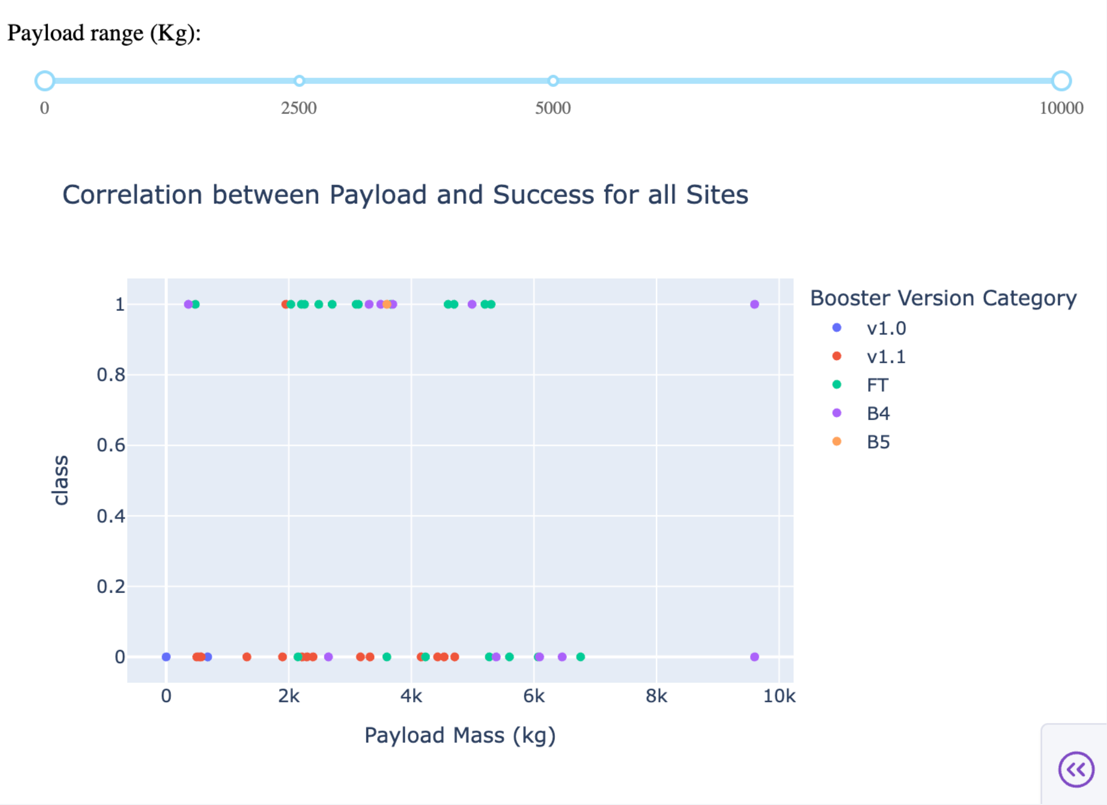
- Explanation:
  - KSC leads in total successes due to frequent use and modern facilities.

Dashboard: Highest Success Site



- Explanation:
  - Highest ratio reflects optimized opeations.

Dashboard: Payload vs. Outcome



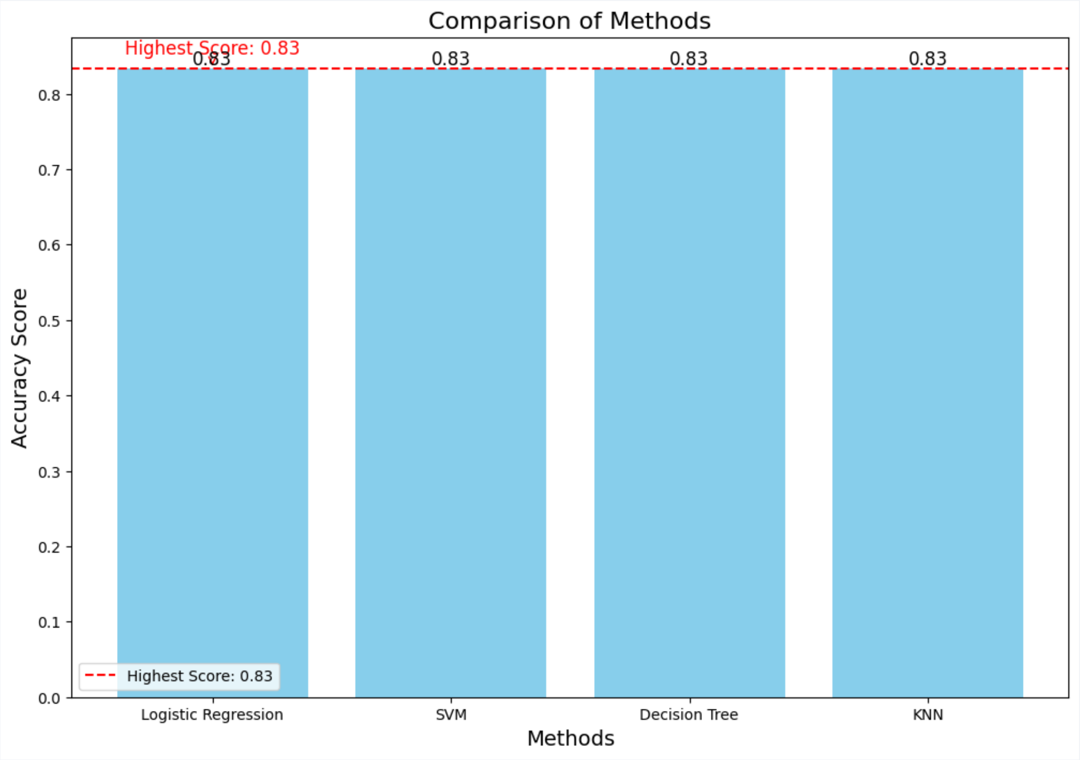
- Explanation:
  - Success declines above 6000 kg;
  - Later booster versions improve rates in mid-range.

Section 5

Predictive Analysis (Classification)

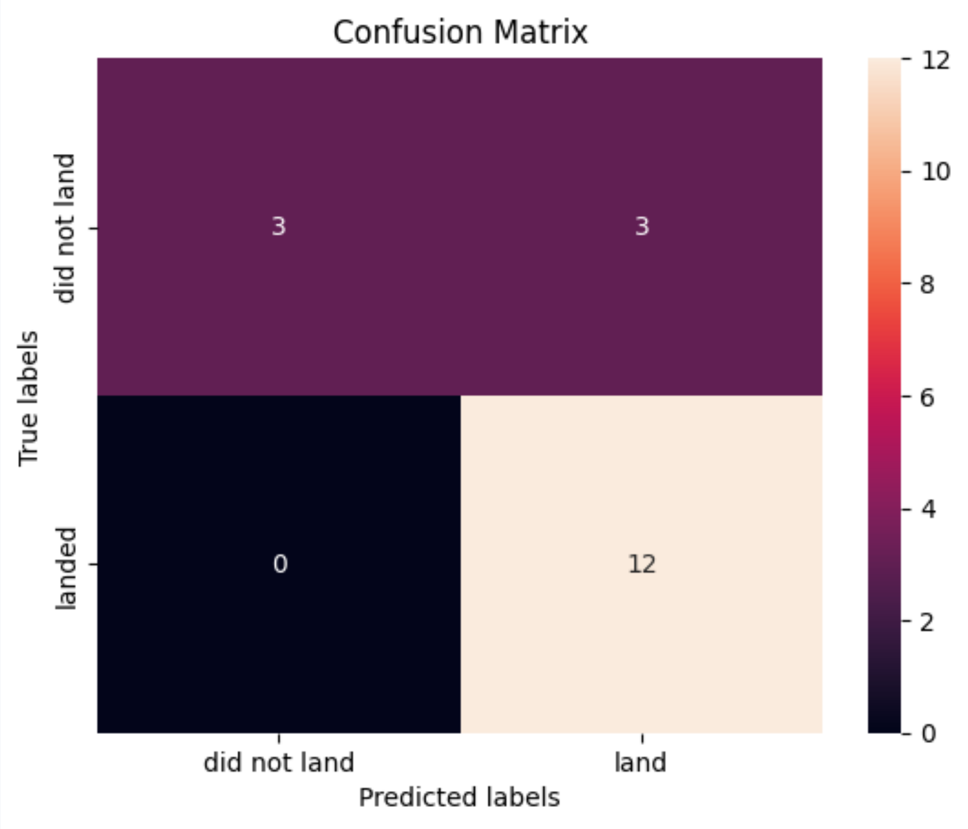
Classification Accuracy

- Bar Chart:
  - Logistic Regression 83%;
  - SVM 83%;
  - Decision Tree 83%;
  - KNN 83% on test set.
- Best Model:
  - Decision Tree (highest cross-validation score; balanced handling of feature interactions)



Confusion Matrix

- Matrix for Best Model:
  - TP 12, TN 3, FP 3, FN 0.
- Explanation:
  - No false negatives minimize risk overestimation;
  - Strong precision for successes.



Conclusions

- **EDA Visualization:** Success trends upward over flights/years; Low payloads and specific orbits (SSO) boost rates.
- **EDA SQL:** Quantified payloads (NASA 45,596 kg) and outcomes (98% success); Historical failures clustered early.
- **Interactive Analytics:** Coastal sites aid landings; Dashboards show 2000-4000 kg as sweet spot.
- **Classification:** 83% accuracy; orbit and payload as top predictors.
- **Innovative Insights:** Booster evolution (FT/B5) correlates with success; proximity analysis suggests safety buffers; potential for ensemble models to push accuracy to 90%+.

Appendix

- Reference:
  - https://www.coursera.org/learn/applied-data-science-capstone/
  - https://github.com/faansing/IBM-Data-Science/

Thank you!